

A Reconfigurable Robot with Lockable Cylindrical Joints

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Abstract—This work presents a new conceptual design for reconfigurable robots. Unlike conventional reconfigurable robots, our design does not achieve reconfigurability by utilizing modular joints. Rather, the robot is equipped with passive joints, i.e., joints without actuator or sensor, which permit changing the Denavit-Hartenberg (DH) parameters such as the link length and twist angle. The passive joints will become controllable when the robot forms a closed kinematic chain. Also, each passive joint is equipped with a built-in brake mechanism which is normally locked, but the lock can be released whenever the parameters are to be changed. Such a versatile and agile robot is particularly suitable for space application for its simple, compact, and light design. The kinematics and recalibration of this kind of reconfigurable robot is thoroughly analyzed. A stable reconfiguration-control algorithm is devised to take the robot from one configuration to another by directly regulating the passive joints to the associated, desired DH parameters. Conditions for the observability and the controllability of the passive joints are also derived in detail.

Index Terms—reconfigurable robots, redundant manipulator, parallel manipulator, constrained robot, space manipulator

I. INTRODUCTION

ROBOTIC manipulators working in extreme environments often need to change their configuration in order to meet the demands of a specific task within the constraints of the environment [1]. Particularly in space applications [2], it is desirable and cost effective to employ a single versatile robot capable of performing different tasks such as inspection, contact operation, assembly (insertion and removal of objects), and carrying objects (pick and place). Optimal operation of each of these tasks demands a specific manipulator design. For instance, large robots maximizing the *structural length index* are typically suitable for inspection [3]; robots with maximum *manipulability measure* are well-conditioned for dexterous contact tasks [4], [5]; and configurations maximizing the distance of the robot limbs and extremities from the environment are suitable for payload handling [6]. The optimization of a robot workspace over its link lengths, as the design parameters, is reported in [7], [8], while optimization of kinematic parameters and criteria for fault tolerance are discussed in [9].

Generally speaking, space systems are designed for minimum weight in order to reduce launch cost. Another design constraint for a space system is that it should be compact

enough to be accommodated within its designated space in the launch vehicle. Since the links of a space manipulator are usually long, they have to be folded before launch. For example, the Canadarm II has two long booms, each of which has a hinge at the middle, which allowed the booms to be folded before launch and then unfolded manually by astronauts in orbit. For on-orbit servicing missions whereby no human operator is present, the robot has to be able to deploy itself.

The original reconfigurable robot was introduced in [10] to add versatility to the robotic manipulator. The concept was then developed further in [11]. Cellular robots based on hexagonal modules and based on the concept of robot molecules were described in [12], [13] and [14], [15], respectively. Reconfigurable robots for space exploration were proposed in [2]. The design of Conro modules to build deployable modular robots that can be reconfigured to take different shapes such as snakes or hexapods were presented in [1]; this work has some similarities with earlier works on Tetrobot, discussed in [16]. Another modular reconfigurable manipulator, which had three 6-DOF arms, was proposed for space applications in [17]–[19]; the manipulator was designed to be able to brachiate around the Japanese section of the International Space Station. All these reconfigurable robots are modular, hence needing an effective docking system for connecting and releasing the modules [20].

Reconfigurable robots have been identified by many space agencies to play a major role in future space missions. In this regard, modular robots have the great advantage of being able to change their both morphology and topology; more specifically, they can change their number of links, form a closed chain, break a chain to a single or multiple branches, and create a tree-like structure, for example. All these capabilities, however, come at the expense of complexity in the joints and their docking system. The new design, on the other hand, can offer a simpler and more effective solution to the problem at hand. Additionally, with such a design, one does not need to detach any link or joint to achieve a required configuration change. Thus, the entire reconfiguration operation can be performed autonomously with a higher level of reliability. This makes this new type of reconfigurable robots particularly attractive for space applications. Further to the versatility that this design provides, it makes it possible to contain the manipulator in a small volume, which is suitable for launch.

In this paper, we describe the conceptual design of a new

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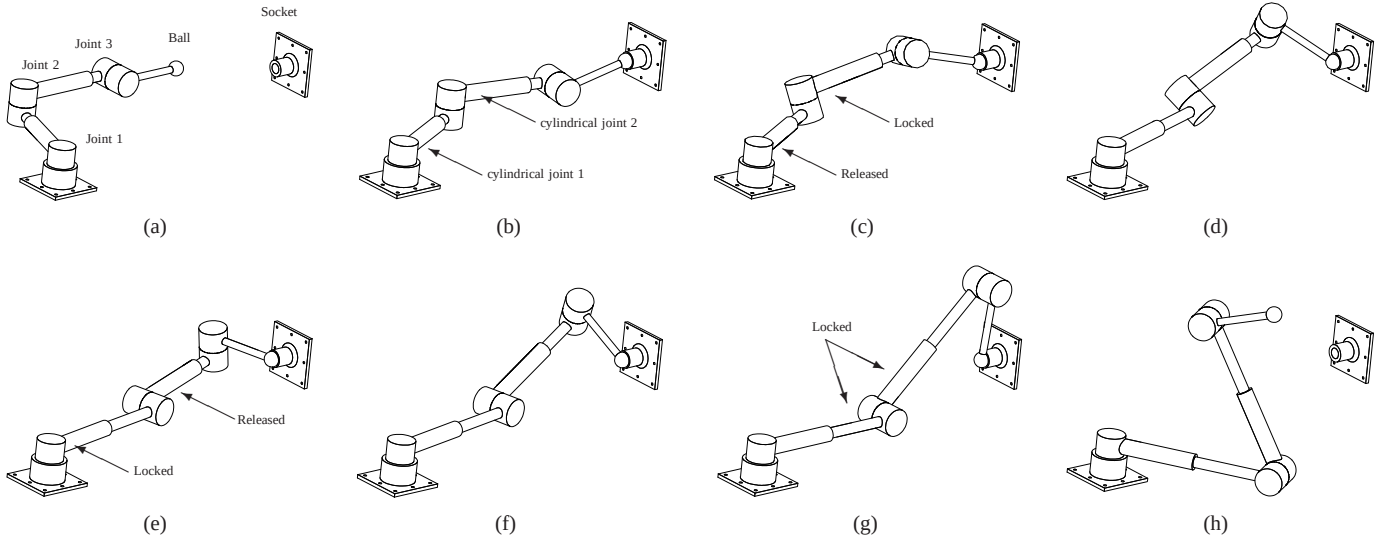


Fig. 1. Reconfiguration maneuver from a short, planar configuration to an anthropomorphic one (grasping device not shown here); (a) the initial configuration; (b) constraining the EE motion with a ball joint; (c,d,e) unlocking and reconfiguring the first passive joint; (f,g) unlocking and reconfiguring the second passive joint; (h) the final configuration.

breed of reconfigurable robots that, using *lockable passive*¹ cylindrical joints (CJ's) between two adjacent *active joints* with noncoinciding axes, can change their kinematic design parameters, i.e., their Denavit-Hartenberg (DH) parameters. Notably, this design does not rely on modular joints. These robots have as many passive cylindrical (telescopic) joints as half the adjustable DH parameters. Each cylindrical joint can be considered as the combination of two passive revolute and prismatic joints. The passive joints do not have any sensors or driving actuators, but they are equipped with a normally locked brake mechanism which can be unlocked simply by activating a solenoid—outlines on preliminary design of such a brake system can be found in [22]–[25]. When locked, the entire cylindrical joint becomes a rigid link which connects the two neighboring active joints. The twist angle and the length of this newly formed link is determined by the values of the locked passive joints.

Schematics provided in Fig. 1 demonstrate the reconfiguration process of a typical reconfigurable manipulator with two cylindrical joints. The kinematics and dynamics aspects of such a robotic operation will be elaborated later in Section V. Here, we only describe the operation concept.

In general, a reconfigurable manipulator can change its configuration by completing the following steps:

- (i) the robot manipulator forms a closed kinematic chain, for example, by grasping a fixed point or, as shown in Fig. 1(b), by the end-effector (EE) holding a ball which is grasped by a socket²;
- (ii) one or more of the brake mechanisms are released,

thus increasing the degrees of freedom (DOF) of the constrained system;

- (iii) the closed-chain system is controlled in such a way that the desired active-joint positions are achieved, Figs. 1(c)–1(d); upon converging to these values, the brakes are locked again;
- (iv) the constraint on the motion of the EE is removed, e.g., by releasing the ball. A manipulator with a new configuration is now born, Fig. 1(h).

Depending on the number of changes needed in the DH parameters, steps (ii) and (iii) may need to be repeated with a few of the passive joints unlocked at a time.

It is worth mentioning that the main advantage of this type of reconfigurable robots is their simplicity. Despite this simplicity, the reconfigurable robot possesses the same dexterity and workspace as a hyper-redundant manipulator with less weight and at a lower cost.

Our strategy in reconfiguration control is based on relatively simple implementability: it only requires knowledge of the robot kinematics and its gravitational torque model. This is clearly an advantage for control implementation, as the complete dynamics model of the constrained system will no longer be needed.

The paper is organized as follows: Firstly, the design concept of the new reconfigurable robots is presented in Section II. Next, we develop the kinematics of the reconfiguration process in Section III. Section IV is devoted to the development of a control methodology for the reconfiguration of a robot with passive joints. Finally, in Section V numerical results obtained by simulating the reconfiguration process illustrated in Fig. 1.

II. DESIGN CONCEPT

The goal is to build reconfigurable robots with the capability to change their kinematic parameters according to the particular demands of a task. Fig. 1 schematically illustrates the steps

¹The idea of using passive joints in a kinematic chain is not new *per se*, as these joints have been used in closed chains such as the well-known four-bar linkage for quite a long time. The effect of having passive joints on the dynamics of some serial chains has also been studied in the context of underactuated serial manipulators; see [21], for example. However, the idea of using lockable, passive cylindrical joints to change the DH parameters of a serial manipulator is entirely novel.

²The grasping mechanism is not shown in the figures for simplicity.

required for reconfiguring an originally short robot into an anthropomorphic robot with long arms. Such a reconfiguration requires changing both twist angles and lengths of the two links as shown in the figure.

A single passive cylindrical joint, which connects two adjacent active joints, provides both translational and rotational motions required for a configuration change. In order to simplify the design, we do not consider any driving actuators or sensors for the passive joints. Also, all components of the locking device are to be installed inside the inner link, resulting in a compact design. The passive joints can be controllable and observable when the robot EE is properly constrained, i.e., the total DOFs of the released passive joints is equal or less than that lost due to the constraint.

It should be pointed out that constraining the robot EE to a fixed position is only one means of forming a closed kinematic chain; there are others. For example, Fig. 2 shows a dual-arm space manipulator which forms the closed chain by connecting the EEs of the two arms. The sub-figures illustrate how the arms are deployed from their packing configurations through different steps. The steps are basically the same as those shown in Fig. 1. The only difference is that, here, the closed kinematic chain—required for the configuration change—is formed by two arms grasping a common object rather than by each arm grasping a fixed point. Fig. 2(a) shows the two arms in their launch configuration, which is planar, of minimum boom length, and positioned in a compact form. That is followed by Fig. 2(b), which shows the departure from the initial postures. Figs. 2(c)–2(e) depict three snapshots of the sequence of events leading to the reconfiguration of both arms. Finally, Fig 2(f) illustrates the two arms in their desired configuration with long booms.

III. RECONFIGURATION KINEMATICS

As mentioned before, the reconfigurable mechanical manipulators proposed in this paper can achieve reconfigurability only by simultaneously releasing one or more of their *passive* joints and constraining the EE motion, the latter of which makes the originally open kinematic chain closed. The motion constraints are then used as a tool to change the joint variables of the released passive joints, which are in fact among the DH parameters of the manipulator. The number of the parameters that can be changed at the same time depends on such kinematic properties as the manipulator degrees of freedom, the dimension of the EE constraint space, and the number of cylindrical joints released. The first parameter is known for each manipulator; the second can be considered a design parameter of the manipulator; the last, however, is case dependent and may need to be determined for each reconfiguration maneuver based on the other kinematic parameters of the manipulator. This is discussed further in Subsection III-B.

A. The Constraint Equations

Assuming that there are n active joints and m released passive, cylindrical joints, we denote the active and the passive generalized coordinates of the manipulator by $\theta \in \mathbb{R}^n$ and $\psi \in \mathbb{R}^{2m}$, respectively. Note that ψ accounts only for the

released passive joints; naturally, the total number of the locked and unlocked joints is higher than or equal to m . The independent external motion constraints of the end-effector are then described by the r -dimensional vector function ϕ

$$\phi \equiv \phi(\theta, \psi) = 0, \quad (1)$$

the number of DOFs of the closed chain thus being

$$d = n + 2m - r \quad (2)$$

Differentiating (1) with respect to time, we obtain

$$A_\theta \dot{\theta} + A_\psi \dot{\psi} = 0, \quad (3)$$

where $A_\theta = \partial\phi/\partial\theta$ and $A_\psi = \partial\phi/\partial\psi$; unless $n = r$ or $2m = r$, neither of these two matrices is square. The above equation can be written in a more compact form as

$$A\dot{q} = 0, \quad (4)$$

where the vector of generalized coordinates q and the Jacobian matrix $A \in \mathbb{R}^{r \times (n+2m)}$ of the constraint equation (1) are defined as

$$q \triangleq \begin{bmatrix} \theta \\ \psi \end{bmatrix}, \quad (5)$$

$$A \equiv [A_\theta \quad A_\psi] \triangleq \frac{\partial\phi}{\partial q}. \quad (6)$$

Now, let us assume the following holds:

Assumption 1: The Jacobian A_ψ remains full-rank during a reconfiguration maneuver.

Then, by making use of (3), one can *uniquely* compute the value of $\dot{\psi}$ from the measured value of $\dot{\theta}$ through

$$\dot{\psi} = Q\dot{\theta}, \quad \text{where } Q \triangleq -A_\psi^+ A_\theta, \quad (7)$$

with superscript $+$ denoting the pseudo-inverse. Note that the matrix $A_\psi \in \mathbb{R}^{r \times 2m}$ being full-rank requires³ that

$$r \geq 2m. \quad (8)$$

This means that, in order to be able to observe the value of ψ , the number of passive DOFs should not exceed the number of constraints.

B. The Number of Released Joints

The control method proposed in Section IV-B works only if there exist adequate numbers of active joints and independent constraints. Since the occurrence of singularities in constraints increases the number of DOFs, the system under the control law may no longer be controllable if the number of DOFs exceeds the number of actuated joints. On the other hand, having too many constraints may reduce the number of DOFs to the extent that it would not be possible to have enough independent passive DOFs.

Therefore, the natural questions are that, for a given robot with n active joints, how many cylindrical joints—namely, m —are allowed to be released at a time and how many independent EE constraints—namely, r —are required? Below,

³Apparently, this is the necessary condition. The sufficient condition for the important case of $m = 1$ when the EE position is constrained is discussed in Subsection III-C.

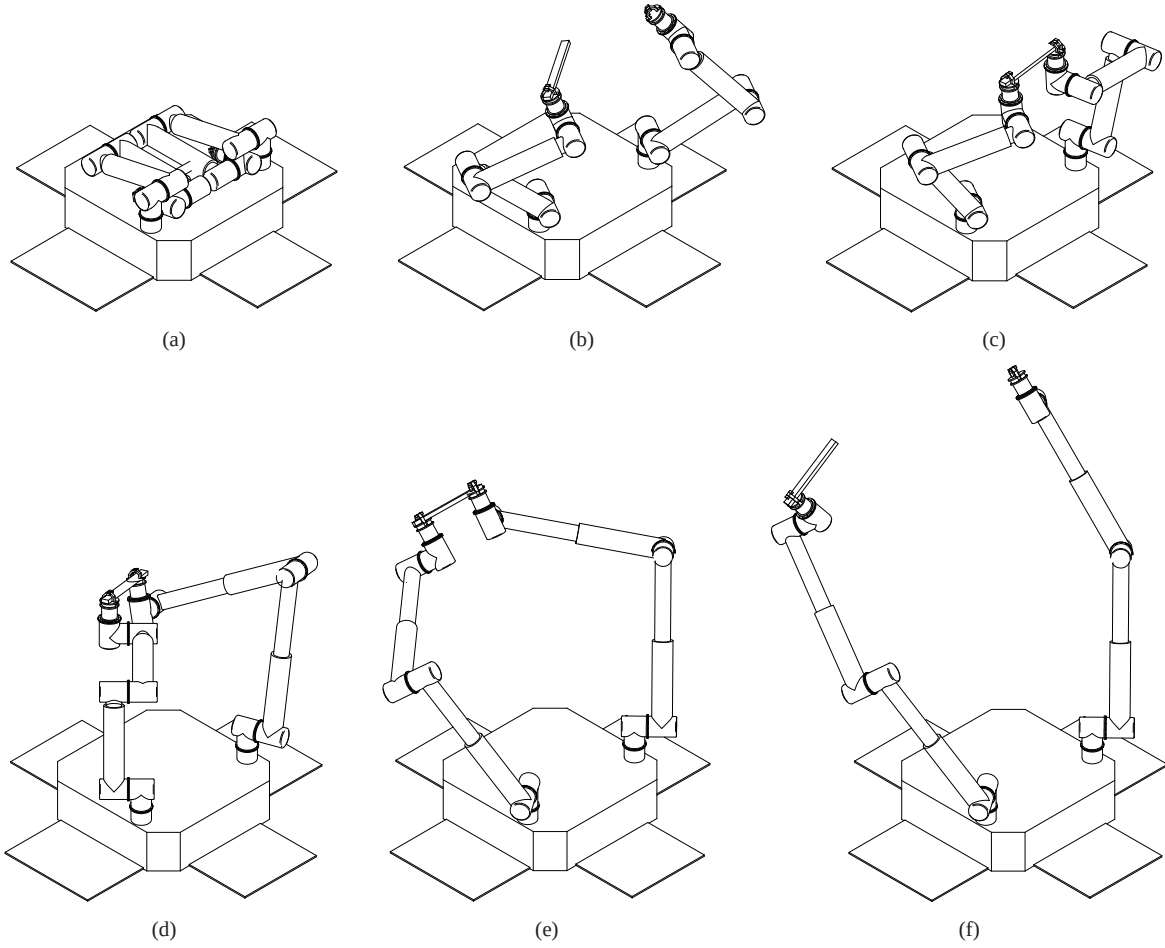


Fig. 2. Reconfiguration maneuver of a dual-arm robot mounted on a servicing satellite; (a) the arms are in the planar configuration and folded for a maximum packing; (b) the planar arms are unfolded; (c) the closed-chain kinematic is formed by the two arms grasping a bar; (d) the left arm reconfigures itself to a special configuration with long booms; (e) the right arm reconfigures itself as the other one; (f) the arms disconnect.

we will show the relation between n , m , and r for a feasible configuration change.

As mentioned in the previous section, to control the constrained robot, we avoid having an underactuated system by releasing only a number of the passive joints. In general, a mechanical system becomes underactuated when the number of degrees of freedom of the system exceeds the number of actuators. Thus,

$$d = n + 2m - r \leq n, \quad \text{or} \quad 2m \leq r, \quad (9)$$

which is identical to (8). Moreover, the minimum number of DOFs must be $2m$ for all the $2m$ elements of vector ψ to be independent variables. Therefore, we have $n + 2m - r \leq 2m$, which is equivalent to this simple inequality: $r \leq n$. Combining the latter inequality with (9) and considering that the maximum number of independent motion constraints which can be applied on the EE is six, we obtain

$$2m \leq r \leq \min(n, 6). \quad (10)$$

This equation gives the lower and upper bounds on the number of required constraints.

Apparently, to be able to vary the passive DOFs independently, ψ must form a complete space, i.e., ψ_i 's must be

TABLE I
THE NUMBER OF PASSIVE JOINTS THAT CAN BE RELEASED AT A TIME

n	2	3	4	5	6
m	1	1	1 2	1 2	1 2 3
r	2	2,3	2-4 4	2-5 4-5	2-6 4-6 6

independent and span the entire $\mathbb{R}^{2m}$. This fact in conjunction with the mapping $Q : \mathbb{R}^n \rightarrow \mathbb{R}^{2m}$ in (7) implies that $2m \leq n$, which is equivalent to

$$m \leq \lfloor \frac{n}{2} \rfloor, \quad (11)$$

where $\lfloor \cdot \rfloor$ denotes the *floor* function, producing the integer part of its argument.

Now, for a given robot with n active joints, in the first step, inequality (11) can be used to determine the maximum number m of the cylindrical joints that can be released. Then, with n and m at hand, inequalities (10) can be used to determine the minimum and the maximum numbers of the EE constraints.

As an example, for the manipulator of Fig. 1, n equals three, and there are two cylindrical joints. However, (11) specifies that only $m = 1$ is acceptable, which means that only one cylindrical joint should be released at a time. Then, based

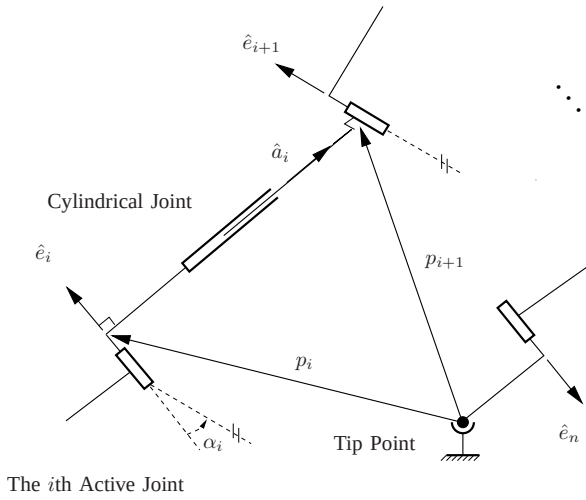


Fig. 3. Axes of the manipulator joints

on (10), r should be greater than or equal to 2 and less than 4; we choose 3, which can be accommodated by constraining the end-effector to a point through a spherical joint, for example.

Different possible topologies satisfying the conditions for a configuration change are given in Table I. Note that often different scenarios for a reconfiguration operation is possible. For example, for a robot with four active joints, namely, $n = 4$, either one cylindrical joint or two cylindrical joints can be released. For the former case, the number of constraints can be two, three, or four; for the latter case, however, the only possibility is four.

C. Singularity Analysis

The condition that the Jacobian matrix A_ψ should remain nonsingular during the configuration change is a critical condition for both observability and controllability of the passive joints. Therefore, the robot should properly be positioned such that the initial and final configurations are not in the vicinity of singularities. There are numerical methods for computing the rank-deficiency locus of a nonsquare Jacobian matrix [26], [27]. For the general case of a reconfigurable robot, it is difficult to identify all postures that render the constraint Jacobian singular. However, we can find the singular postures for the important case where all translational motions of the EE is constrained using a ball-joint type of constraint, as shown in Fig. 3. According to (10), if $r = 3$, the only possibility is that $m = 1$, i.e., only one cylindrical joint can be released at a time. Assuming that the cylindrical joint connects two adjacent active joints i and $i + 1$. Then, the structure of the overall Jacobian comprising the rotary and the prismatic joints takes on this form:

$$A = \underbrace{\begin{bmatrix} p_1 \times \hat{e}_1 & \cdots & p_n \times \hat{e}_n \end{bmatrix}}_{A_\theta} \underbrace{\begin{bmatrix} \hat{a}_i & p_i \times \hat{a}_i \end{bmatrix}}_{A_\psi} \quad (12)$$

where $\hat{e}_i$'s are unit vectors associated to the active-joint axes; p_i 's are the vectors connecting the fixed point to the joints; and the unit vector $\hat{a}_i$ lies along the axis of the cylindrical

joint, which is also the common normal of $\hat{e}_i$ and $\hat{e}_{i+1}$, i.e.,

$$\hat{a}_i = \frac{p_{i+1} - p_i}{\|p_{i+1} - p_i\|}. \quad (13)$$

Thus

$$A_\psi = \kappa_i \begin{bmatrix} p_{i+1} - p_i & p_i \times p_{i+1} \end{bmatrix}, \quad (14)$$

where $\kappa_i = 1/\|p_{i+1} - p_i\|$. Note that submatrix A_θ is essentially a block of the Jacobian matrix of a serial manipulator pertaining to the manipulator translational motion⁴.

It is apparent from (12) that the two columns of A_ψ are perpendicular to one another, and that submatrix A_ψ becomes rank deficient only when $p_i \times \hat{a}_i = 0$, or, equivalently, $p_i \times p_{i+1} = 0$. The loci of the rank deficient configurations can then be expressed as

$$\mathcal{D} = \{q \in \mathbb{R}^{n+2m} \mid \phi(q) = 0 \quad p_i(q) \times p_{i+1}(q) = 0\}. \quad (15)$$

This means that A_ψ can only become singular when the moment of the passive-joint axis about the fixed point vanishes. Therefore, in order to satisfy the condition of Corollary 1, one may consider the following remark.

Remark 1: When the manipulator is reconfigured by constraining only the translational motion of its EE (e.g., Fig. 1), Assumption 1 will hold if the fixed point or the EE does not lie on the axis of the released cylindrical joint.

By inspection, one can show that, for the system of Fig 1, the condition of Remark 1 for the first cylindrical joint will always be satisfied if the socket height does not match the joint offset, while, for the second cylindrical joint, the last link offset has to simply be nonzero.

In the event that the lockable joint has only one DOF, the Jacobian takes the form of either $A_\psi = [p_{i+1} - p_i]/\|p_{i+1} - p_i\|$ or $A_\psi = [p_{i+1} \times p_i]/\|p_{i+1} - p_i\|$ corresponding to a prismatic or a rotary joint, respectively. Clearly, the Jacobian remains always full-rank for the case of prismatic joint. For the case of rotary joint, however, the Jacobian is full-rank if the condition in Remark 1 holds.

We should note here that, when the reconfigurable robot forms a closed kinematic chain, in essence, it becomes a parallel manipulator. The subject of parallel manipulators has been researched by numerous authors, and there is a vast literature on the subject [5]. The subject of singularity, for example, has been studied by many researchers, e.g. [28], [29]. Although the results reported in the literature may not be directly applicable to our problem,⁵ they will indeed be illuminating in further research on reconfigurable robots with passive joints.

⁴The manipulator Jacobian takes on the form

$$J = \begin{bmatrix} p_1 \times \hat{e}_1 & p_2 \times \hat{e}_2 & \cdots & p_n \times \hat{e}_n \\ \hat{e}_1 & \hat{e}_2 & \cdots & \hat{e}_n \end{bmatrix}.$$

⁵One major difference is that, in the case of parallel manipulators, the output of the manipulator is considered to be the motion of a particular link of the manipulator. In the case of a reconfigurable robot, however, of main interest is the value of the unactuated joints.

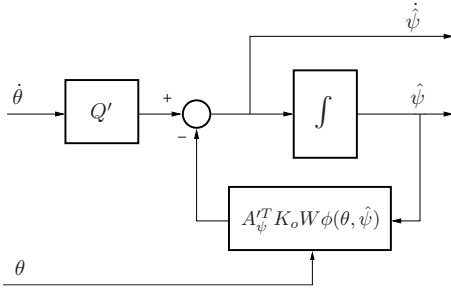


Fig. 4. Estimating the states of the passive joints

D. The Manipulator Recalibration

The purpose of reconfiguring a manipulator is to change some of its DH parameters which are represented by the passive-joint variables. Since the passive joints are not instrumented, after any reconfiguration, the manipulator must be recalibrated; that can be done by computing the passive-joint variables from the constraint equations, given the sensor readings of the active joints. In this sense, the problem of recalibrating a reconfigurable robot is different from that of calibrating a parallel robot, which has been extensively discussed in the literature; for references, see [5]. For the purpose of this paper, recalibration means determining the new values of the passive-joint variables after they are changed through a reconfiguration maneuver while all other kinematic parameters of the manipulator are either already known, e.g., the rest of the DH parameters, or are directly measured, as is the case of active joints.

Taking ψ as the output of the closed chain, we can see a resemblance between the problem of recalibrating a reconfigurable robot and that of solving the direct kinematics of a parallel manipulator, which is a difficult problem to solve due to multiplicity of the solutions [5]. Although the set of closed-form solutions may be found for a given structure, there is no such solution for a general closed chain, i.e., the problem has to be solved on a case by case basis, which is not suitable for the general framework we intend to establish in this paper.

We have developed two numerical algorithms to find the solutions. These algorithms are based on the conjecture that, starting from a known value and avoiding the singularities in which the different branches of the direct kinematics solution meet, we should be able to accurately follow the change in ψ and thus obtain its new value when the reconfiguration maneuver ends. The first algorithm can be seen as a state estimator, which we prove to be globally exponentially stable, and the second is the application of the Newton-Raphson method in solving the constraint equations.

1) *Dynamic Estimator:* Having computed $\dot{\psi}$ from (7), one can then integrate it to obtain the time-history of the passive-joint variables during a reconfiguration maneuver. The integration, however, will inevitably lead to a drift in the position error. In order to suppress this drift, we use a dynamic estimator which employs the constraint equations ϕ as a measure of the estimation error. For convenience, it would be easier to use the same number of “measurements” as the number of variables to be estimated. Therefore, we choose a

combination $\phi' \in \mathbb{R}^{2m}$ of the constraint equations defined by

$$\phi' \triangleq W\phi,$$

where W is a $2m \times r$ full-rank constant matrix; in its simplest form, W can be chosen as a selection matrix that picks $2m$ of the r constraints. Note that (2) implies that, if $n = d$, then $r = 2m$ and the weight matrix can hence be conveniently selected as the identity matrix, i.e., $W = I$. Now, premultiplying the velocity constraint (3) by W , we obtain

$$A_{\theta}' \dot{\theta} + A_{\psi}' \dot{\psi} = 0,$$

in which $A_{\theta}' \triangleq W A_{\theta}$ and $A_{\psi}' \triangleq W A_{\psi}$. Equation (7) can then be simplified as

$$\dot{\psi} = Q' \dot{\theta}, \quad \text{where} \quad Q' \triangleq A_{\psi}'^{-1} A_{\theta}'. \quad (16)$$

As shown in Fig. 4, we realize the dynamic estimator by closing the loop using an $A_{\psi}'^T K_O \phi'$ feedback. This feedback and the feedforward given by (16) result in

$$\dot{\hat{\psi}} = Q'(\theta, \hat{\psi}) \dot{\theta} - A_{\psi}'^T(\theta, \hat{\psi}) K_O W \phi(\theta, \hat{\psi}). \quad (17)$$

In the above equation, $\hat{\psi}$ denotes the estimate of the value of the passive joints, and $K_O \in \mathbb{R}^{2m \times 2m}$ is the positive-definite estimator gain matrix. The inputs and outputs of the estimator loop are $\{\theta, \dot{\theta}\}$ and $\{\hat{\psi}, \dot{\hat{\psi}}\}$, respectively.

Proposition 1: Let us assume that A_{ψ} remains a full-rank matrix during the estimation process, and that an estimate of ψ is obtained from system (17). Then, the constraint equation $\phi(\theta, \hat{\psi})$ as a function of the estimated values of the passive joints globally uniformly asymptotically converges to zero. The estimator will also be globally exponentially stable.

Proof: Consider the positive-definite Lyapunov function candidate

$$V = \frac{1}{2} \phi'^T K_O \phi',$$

which satisfies the following bounds

$$\lambda_{\min}(K_O) \|\phi'\|^2 \leq V \leq \lambda_{\max}(K_O) \|\phi'\|^2. \quad (18)$$

Differentiating V with respect to time along the trajectories of (17) yields

$$\begin{aligned} \dot{V} &= \phi'^T K_O A_{\theta}' \dot{\theta} + \phi'^T K_O A_{\psi}' \dot{\hat{\psi}} \\ &= \phi'^T K_O A_{\theta}' \dot{\theta} - \phi'^T K_O A_{\psi}' (A_{\psi}'^T K_O \phi' + A_{\psi}'^{-1} A_{\theta}' \dot{\theta}) \\ &= -\phi'^T K_O A_{\psi}' A_{\psi}'^T K_O \phi' \\ &\leq -\lambda_{\min}(A_{\psi}' A_{\psi}'^T) \lambda_{\min}^2(K_O) \|\phi'\|^2. \end{aligned} \quad (19)$$

Therefore, based on the Lyapunov stability theory for non-autonomous system [30, p. 138], it can be inferred from (18) and (19) that $\phi'(\psi, \theta) = 0$ must be a globally uniformly asymptotically stable equilibrium point of the system (17). This means that, as t becomes large, ϕ' approaches zero. Consequently, $\hat{\psi}$ will approach the actual value ψ , and ϕ will asymptotically vanish. Furthermore, because the bounding functions of V and $\dot{V}$ are of the form $a \|\phi'\|^b$ where a and b are strictly positive constants, the system is also globally exponentially stable [30, p. 140]. ■

Apparently, the estimator (17) is similar to the closed-loop inverse kinematics (CLIK) scheme [31], [32], where the

inverse kinematics problem is solved by reformulating it in terms of the convergence of an equivalent feedback control system. It should be noted that, whereas the conventional CLIK algorithm [31], [32] only confines the constraint error inside a small ball, our estimator asymptotically eliminates it.

2) *The Newton-Raphson Method*: Alternatively, given the value of the active joints, we can solve the constraint equation for the value of the passive joints using a numerical scheme such as Newton-Raphson's. To this end, the constraint equation (1) is expanded using the Taylor series:

$$\phi(\theta, \psi) = \phi(\theta, \psi_i) + A_\psi(\psi - \psi_i) + \text{H.O.T.} = 0. \quad (20)$$

In the above equation, ψ_i is the value of the passive joints at the i th iteration. If the higher-order terms (H.O.T.) are neglected, we will have

$$\psi_{i+1} = \psi_i - A_\psi^+ \phi(\theta, \psi_i). \quad (21)$$

At any time-step, one will need an initial guess to start the iteration. The value of the passive joints at the previous time-step can be used as this initial guess. Moreover, the iterations can continue until the constraint error become sufficiently small. Note that, here again, A_ψ needs to be of full rank, so that the passive-joint variables can be determined unambiguously.

It should be noted that, when r is strictly larger than $2m$, we have more equations than unknowns, which amounts to having redundant measurements that would help reduce the effect of uncertainties. The drawback, however, is that the algorithm is iterative.

IV. RECONFIGURATION CONTROL

When the passive joints of the manipulator are locked, and the EE is not constrained, the manipulator is controlled using a separate controller—any appropriate controller designed for serial manipulators can be used for that purpose. Since the robot will have some unactuated joints during any reconfiguration maneuver, the main challenge will lie in developing a dedicated controller for reconfiguration.

Control algorithms reported in the literature for closed kinematic chains can be classified in two categories: Those which are not based on the dynamics model of the system, and the ones that are, [33]. What makes the problem of controller design here more challenging is that, if the number of constraints, r , is larger than the number of passive-joint coordinates, $2m$, the system will become overactuated. On the other hand, when $r = 2m$, the system has as many controllable active joints as DOFs. Therefore, the reconfiguration controller must be able to handle both these cases. Three controllers that took into account the overactuation were reported in [34] for redundantly actuated parallel manipulators. Two of these controllers are model-based, and the other is a simple PD that minimizes the norm of the actuation torque. However, the stability of these controllers was not analyzed.

In this section, we devise a gravity-compensated PD projection-based control which is proven to be asymptotically stable. This controller is similar to the stable PD controller developed in [35]. However, there are three major differences: (i) the control loop is directly closed on the value of the unactuated joints, which are to be changed through reconfiguration,

although the actual measurements are on the active joints; (ii) The controller reported here works whether the system is overactuated or not; (iii) one does not need to derive the minimal-order dynamics model of the system. Moreover, the derivation of the projection-based controller seems to be much simpler than the Lagrange-D'Alembert formulation [34], [36] because the structure of the former controller can be obtained from the same vectors of which constitutes the Jacobian matrix of the serial manipulator.

An assumption here is that the actuator of the active joints can deliver torque accurately as requested by the control system. Note that there are numerous works in the literature dealing with accurate control of the joint torque [37]–[42] or the compensation of the joint friction [43]–[45] in robots.

A. The Dynamics Model

After the reconfigurable arm forms a closed chain, one or more of its passive joints are unlocked. The full-order dynamics equations of the mechanical system can then be derived⁶ as

$$M\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau - \tau^c \quad (22)$$

which apparently is subject to the constraint equation

$$\phi(q) = 0, \quad (23)$$

where vector $\tau \in \mathbb{R}^{n+2m}$ represents the generalized force in the active and passive joints, with its first n entries corresponding to the active joints and the rest to the passive ones; τ^c represents the generalized constraint force associated to the Lagrange multipliers; $q \in \mathbb{R}^{n+2m}$ is the vector of generalized coordinates as given in (5); $M(q)$ is the $(n+2m) \times (n+2m)$ inertia matrix; $C(q, \dot{q})\dot{q}$ is the $(n+2m)$ -dimensional vector containing the Coriolis and centrifugal terms; and $g(q) \in \mathbb{R}^{n+2m}$ is the gravity torque.

Now, given the $r \times (n+2m)$ constraint Jacobian matrix A , we can uniquely define symmetric matrix P , the *null-space orthogonal projector* of A , as [46]

$$P \triangleq I - A^+ A \quad (24)$$

where I is the $(n+2m) \times (n+2m)$ identity matrix. Because P is an orthogonal projection onto the null-space of the Jacobian—also known as the *tangent space* of the constraint manifold—any vector in $\mathcal{N}(A)$ is projected onto itself, whereas any vector perpendicular to the tangent space lies in the null-space of P . The vector $\dot{q}$ of generalized velocities belongs to the former group as $A\dot{q} = 0$, and the vector of constraint generalized forces τ^c belongs to the latter because $\forall \dot{q} \in \mathcal{R}(P) \equiv \mathcal{N}(A)$, $\dot{q}^T \tau^c \equiv 0$. In other words, these two relations hold:

$$P\dot{q} = P^T \dot{q} = \dot{q}, \quad \text{and} \quad P\tau^c = 0. \quad (25)$$

⁶Here, any of the multitude of dynamics-modeling methods can be used to derive the equations of motion in the form of (22); the particular method one actually uses to derive these equations is immaterial to the proceeding developments.

Hence, premultiplying (22) by P , one can eliminate τ^c from the set of equations:

$$PM\ddot{q} = P(\tau - C(q, \dot{q})\dot{q} - g(q)). \quad (26)$$

Moreover, the vector of generalized forces τ can be decomposed into two components denoted by subscripts $\parallel$ and $\perp$, lying in the orthogonal subspaces the tangent space $\mathcal{N}(A)$ and the null-space of P , respectively:

$$\tau = \tau_{\parallel} + \tau_{\perp}. \quad (27)$$

Because $\tau_{\perp} \in \mathcal{N}^{\perp}(A)$ and the constrained motion occurs in $\mathcal{N}(A)$, by definition, this component of the actuation generalized forces does not contribute to the motion of the system [47].

To keep the actuation torque to a minimum, the control law reported in Subsection IV-B1 specifies a control torque $\tau_{\parallel}$ that lies in the constraint tangent plane. However, the control torque thus computed may not actually be realizable in the system as it may need actuation in the unactuated joints, i.e., $\tau_{\parallel}$ may have non-zero values among its last $2m$ entries. Therefore, in Subsection IV-B2, we compute a complementary control torque $\tau_{\perp}$ in the null-space orthogonal such that the total control torque, namely, $\tau_{\parallel} + \tau_{\perp}$, will have only zeros as its last $2m$ entries, the torque thus becoming applicable.

B. The Control Law

The control objective is to regulate the passive joints of the system to their desired values ψ_d . The number of independent generalized coordinates of the system is $d = n + 2m - r$, the DOF of the system. Therefore, one can control the constrained mechanical system by controlling only an independent, complete subset $\chi(q) \in \mathbb{R}^d$ of the generalized coordinates.

Assumption 2: The vector of passive-joint variables ψ constitutes an independent set.

The kinematic relation (3) implies that the above assumption is equivalent to

$$\mathcal{R}(A_{\psi}) \subseteq \mathcal{R}(A_{\theta}). \quad (28)$$

Then, $\chi(q)$ can be selected as

$$\chi \triangleq \begin{bmatrix} \psi \\ \eta(\theta) \end{bmatrix},$$

where the elements of vector $\eta(\theta) \in \mathbb{R}^{n-r}$ can be any independent functions of θ . Therefore, the addition of the auxiliary variables η makes χ a complete set of independent variables. Clearly, in the case that the number of EE constraints is equal to that of active joints, namely, $n = r$, the auxiliary variable η vanishes and hence $\chi = \psi$; this is the case for the reconfigurable manipulator of Fig. 1.

The time derivative of χ , which constitutes an independent set of generalized velocities, can then be obtained from

$$\dot{\chi} \equiv D\dot{q} = DP\dot{q} \quad (29)$$

where

$$D \triangleq \frac{\partial \chi}{\partial q} = \begin{bmatrix} 0_{2m \times n} & I_{2m \times 2m} \\ \partial \eta / \partial \theta & 0_{(n-k) \times 2m} \end{bmatrix}$$

is the Jacobian of the independent variables.

1) *Null-space Control Torque:* Let us consider the following control law:

$$\tau_{\parallel} = -PD^T(K_D\dot{\chi} + K_P(\chi - \chi_d)) + Pg(q) \quad (30)$$

where K_D and K_P are $d \times d$ positive-definite feedback gains, and vector $\chi_d^T = [\psi_d^T \quad \eta_d^T]$ contains the desired variables.

Theorem 1: The constrained mechanical system (22) under the control law (30) asymptotically converges to the desired position χ_d .

Proof: Substituting control law (30) in the dynamics equation (26), we obtain

$$PM\ddot{q} = -PC(q, \dot{q})\dot{q} - PD^TK_D\dot{\chi} - PD^TK_Pe, \quad (31)$$

where $e \triangleq \chi - \chi_d$ is the position error. Now consider the following candidate Lyapunov function:

$$V = \frac{1}{2}\dot{q}^TM\dot{q} + \frac{1}{2}e^TK_Pe, \quad \forall \dot{q} \in \mathcal{N}(A). \quad (32)$$

Then, using (29) and the first of (25), one can compute the time-derivative of the above function along the solution of (31):

$$\begin{aligned} \dot{V} &= \frac{1}{2}\dot{q}^T\dot{M}\dot{q} + \dot{q}^TM\ddot{q} + \dot{\chi}^TK_Pe \\ &= \frac{1}{2}\dot{q}^T\dot{M}\dot{q} + \dot{q}^TPM\ddot{q} + \dot{\chi}^TK_Pe \\ &= \dot{q}^T\left(\frac{1}{2}\dot{M} - C\right)\dot{q} - \dot{\chi}^TK_D\dot{\chi} \end{aligned}$$

However, since C can be chosen such that $\dot{M} - 2C$ is a skew-symmetric matrix [48], we will have

$$\dot{V} = -\dot{\chi}^TK_D\dot{\chi} \leq 0 \quad (33)$$

which is negative-semidefinite. Clearly, we have $\dot{V} = 0$ only if $\dot{\chi} = 0$, or if $\dot{q} = 0$ because there is a one-to-one relationship between $\dot{\chi}$ and $\dot{q}$. Now, substituting $\dot{\chi} = 0$ and $\ddot{q} = \dot{q} = 0$ in (31), we can find the largest invariant set with respect to system (31) as

$$\Omega = \{\chi, \dot{\chi} : \dot{\chi} = 0, \quad PD^TK_P(\chi - \chi_d) = 0\} \quad (34)$$

On the other hand, one can see from (29) that DP —and thus its transpose PD^T —must be a full-rank matrix as χ was selected to be a complete set of independent generalized coordinates. Therefore, the vector equation inside (34) can only hold if $e = \chi - \chi_d$ vanishes. Then, $\Omega = \{\chi = \chi_d, \dot{\chi} = 0\}$ is the largest invariant set which satisfies $\dot{V} = 0$. Therefore, according to LaSalle's Global Invariant Set Theorem [49], [30, p. 115], the solution of system (31) asymptotically converges to the invariant set Ω . Consequently, as the time progresses, χ and thus ψ asymptotically approach their desired values χ_d and ψ_d , respectively. ■

Remark 2: The control torque proposed above is of minimal norm in the sense that it does not contribute to the constraint forces.

2) *Admissible Control Torque*: Because the passive joints are unactuated, the vector of the generalized forces should contain as many zeros as twice the number of the released passive joints. There are n single-DOF active joints and m two-DOF passive joints. Thus, corresponding to the definition of q , the generalized input force will be of this form:

$$\tau = \tau_{\parallel} + \tau_{\perp} = \begin{bmatrix} \tau_{\theta} \\ \tau_{\psi} \end{bmatrix} \quad \text{with} \quad \tau_{\psi} = 0_{2m} \quad (35)$$

Here, vector $\tau_{\theta} \in \mathbb{R}^n$ represents the actuation torque, applied at the active joints, while the zero vector τ_{ψ} is due to the lack of actuation in the passive joints. This implies that any admissible generalized force satisfies

$$B\tau = \tau, \quad \text{and} \quad \tau \in \mathcal{B} \triangleq \mathcal{R}(B), \quad (36)$$

in which orthogonal projection B onto the *actuator space* $\mathcal{B}$ is defined as

$$B \triangleq \begin{bmatrix} I_{n \times n} & 0_{n \times 2m} \\ 0_{2m \times n} & 0_{2m \times 2m} \end{bmatrix}. \quad (37)$$

As mentioned before, the control law (30) produces control torques that are completely in the tangent space. However, these torques may not necessarily be admissible, meaning that they may not lie in the range of B . Therefore, we need to modify the motion control law (30) so as to fulfill the condition in (36). If $\mathcal{N}(A) \subseteq \mathcal{R}(B)$, then (36) is automatically satisfied by choosing $\tau = \tau_{\parallel}$, where $\tau_{\parallel}$ is obtained from (30). Otherwise, we need to add an $\mathcal{N}^{\perp}(A)$ component, say $\tau_{\perp}$, to $\tau_{\parallel}$ so that $\tau = \tau_{\parallel} + \tau_{\perp}$ lie in $\mathcal{B}$. Note that, since $\tau_{\perp}$ does not affect the system motion at all, the motion tracking performance of the controller is preserved.

Premultiplying both sides of (35) by P , we arrive at

$$\tau_{\parallel} = \begin{bmatrix} P_1 & P_2 \end{bmatrix} \begin{bmatrix} \tau_{\theta} \\ 0_{2m} \end{bmatrix} = P_1 \tau_{\text{act}}, \quad (38)$$

in which the projection matrix has been partitioned into submatrices $P_1 \in \mathbb{R}^{(n+2m) \times n}$ and $P_2 \in \mathbb{R}^{(n+2m) \times 2m}$. Given $\tau_{\parallel}$, the above equation will have at least one solution for τ_{act} if

$$\mathcal{N}(A) \subseteq \mathcal{R}(P_1). \quad (39)$$

In that case, there is a τ_{act} that can produce the generalized torque control $\tau_{\parallel}$. The minimum-norm solution, i.e., $\|\tau_{\text{act}}\| \rightarrow \min$, can be obtained using the pseudo-inverse of P_1 :

$$\tau_{\theta} = P_1^+ \tau_{\parallel}. \quad (40)$$

Finally, substituting $\tau_{\parallel}$ from (30) into (40), we can derive the motor-torque control law as

$$\tau_{\theta} = -P_1^+ P D^T (K_D \dot{\chi} + K_P (\chi - \chi_d)) + P_1^+ P g(q). \quad (41)$$

Remark 3: The two submatrices P_1 and P_2 of the projection matrix P can be used to select the $\dot{\theta}$ and $\dot{\psi}$ parts of $\dot{q}$:

$$\dot{\theta} = P_1^T \dot{q} \quad (42)$$

$$\dot{\psi} = P_2^T \dot{q} \quad (43)$$

Remark 4: From (43), it is clear that Assumption 2, i.e., the requirement for the entries of $\dot{\psi}$ to be independent, implies that P_2 should be of full rank.

C. Kinematic Conditions for Controllability

Condition (39) may seem too restrictive or difficult to satisfy, especially that one cannot easily manipulate either of the two subspaces involved. However, this concern is a nonissue. In fact, we can show that, if Assumption 1 holds, (39) is automatically satisfied. In other words, if the values of the passive joints can be uniquely determined from those of the active joints, then they can also be changed to their desired values.

Proposition 2: If the Jacobian matrix A_{ψ} is full-rank, then

i) there is no nonzero vector that lies in both $\mathcal{N}(A)$ and $\mathcal{N}(P_1^T)$:

$$\mathcal{N}(P_1^T) \cap \mathcal{N}(A) = \emptyset \quad (44)$$

ii) the range of P_1 is the same as the null-space of the constraint Jacobian:

$$\mathcal{R}(P_1) = \mathcal{N}(A). \quad (45)$$

Proof: We prove the first part of the proposition by contradiction. To this end, let us consider a vector $\xi \neq 0$ that lies in both $\mathcal{N}(A)$ and $\mathcal{N}(P_1^T)$. Then, by definition, ξ must satisfy both

$$A\xi = 0 \quad \text{and} \quad P_1^T \xi = 0. \quad (46)$$

The first relation is the same as (3); as such, one can divide ξ into two sub-arrays $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^{2m}$ —corresponding to $\dot{\theta}$ and $\dot{\psi}$, respectively—such that $\xi^T = [u^T \ v^T]$. Moreover, if A_{ψ} is full-rank, one can compute v from u using a relation similar to (7): $v = Qu$. Then, comparing the second of (46) with (42), we can see that

$$\begin{aligned} u = P_1^T \xi = 0 & \Rightarrow v = Qu = 0 \\ \Rightarrow \xi \equiv \begin{bmatrix} u \\ v \end{bmatrix} &= 0, \end{aligned}$$

which is a contradiction, i.e., the only vector ξ that satisfies (46) is the zero vector, the trivial solution. That completes the proof of the first part of the proposition.

For the second part, we notice that (44) amounts to

$$\mathcal{N}(A) \subseteq \mathcal{N}^{\perp}(P_1^T). \quad (47)$$

To relate the above relation to the range of P_1 , we resort to the fundamental theorem of linear operator transformation, which states that the range of a linear operator is the same as the null-space orthogonal of its transpose.⁷ For P_1 , this means

$$\mathcal{R}(P_1) = \mathcal{N}^{\perp}(P_1^T) \quad (48)$$

which combined with (47) results in $\mathcal{N}(A) \subseteq \mathcal{R}(P_1)$. However, $\mathcal{R}(P_1)$ is evidently a subset of $\mathcal{R}(P) \equiv \mathcal{N}(A)$. Hence, we must have

$$\mathcal{N}(A) \subseteq \mathcal{R}(P_1) \subseteq \mathcal{N}(A) \Rightarrow \mathcal{R}(P_1) = \mathcal{N}(A),$$

which completes the proof. ■

The results of the above development can be briefly stated as follows:

⁷The fundamental relationship between the null-space and the range space of a linear transformation S is $\mathcal{R}(S^T) = \mathcal{N}^{\perp}(S)$ [46], [50].

Corollary 1: Assume that the followings hold:

$$A_\psi \text{ is full-rank} \quad (49a)$$

$$\mathcal{R}(A_\psi) \subseteq \mathcal{R}(A_\theta) \quad (49b)$$

Then,

- (i) the states of the passive joints can uniquely be obtained from those of the active joints, e.g., equations (7) or (17), and
- (ii) the torque-control law (41) forces the actual positions of the passive joints, ψ , and the auxiliary variable, η , asymptotically converges to their desired values ψ_d and η_d , while demanding minimum actuation force.

D. Unattainable Configurations

For the case of reconfigurable robots with a ball-socket type of constraint, we have already shown that A_ψ can become a full-rank matrix under a mild condition. Therefore, the reconfigurability rests on condition (49b), where

$$\mathcal{R}(A_\theta) = \text{span}\{p_1 \times \hat{e}_1, \dots, p_n \times \hat{e}_n\} \quad (50a)$$

$$\mathcal{R}(A_\psi) = \text{span}\{p_{i+1} - p_i, p_{i+1} \times p_i\} \quad (50b)$$

In this section, we find the relationship between the robot axes and condition (49b) that can be helpful in the identification of attainable and unattainable configurations. First, as an illustration of an unattainable configuration, we will show in the followings that a reconfigurable robot with constraint arrangement of Fig. 1 cannot be reconfigured to or from a planar configuration, where all joint axes are parallel.

1) *Parallel axes:* If all joint axes are parallel, i.e., $\hat{e}_1 = \dots = \hat{e}_n$, then $\hat{e}_1 \notin \mathcal{R}(A_\theta)$ because

$$\hat{e}_1 \cdot (p_i \times \hat{e}_i) = 0 \quad i = 1, \dots, n.$$

On the other hand, if the axis vector $\hat{e}_1$ does not lie in the plane of vectors p_i and p_{i+1} , we can say

$$\hat{e}_1 \cdot (p_{i+1} \times p_i) \neq 0,$$

meaning that $\mathcal{R}(A_\psi)$ has a non-zero component along $\hat{e}_1$; hence, $\mathcal{R}(A_\psi) \not\subseteq \mathcal{R}(A_\theta)$.

2) *Testing the reconfigurability condition:* To ascertain the feasibility of a particular configuration change, the initial and final robot positions have to be checked for condition (49b). Since $\text{rank}(A_\psi) = 2$, the latter condition cannot be satisfied if all vectors in set (50a) are parallel. Moreover, if the rank of A_θ is 3, the condition (49b) is trivially satisfied. Therefore, the tricky case is when only two vectors in the set are not parallel demanding identical range spaces in (50). In view of the relationship between the range and the null space of a linear transformation, condition (49b) is equivalent to $\mathcal{N}(A_\theta^T) \subseteq \mathcal{N}(A_\psi^T)$. On the other hand, the null space of A_ψ^T is any vector lying on the plane of p_{i+1} and p_i while being perpendicular to $\hat{a}_i \propto p_{i+1} - p_i$, i.e., any vector aligned with $(p_{i+1} - p_i) \times (p_{i+1} \times p_i)$. Therefore, in the event that only two nonparallel vectors exist in (50a), the following equations have to be satisfied:

$$(p_{i+1} - p_i) \times (p_{i+1} \times p_i) \cdot (p_j \times \hat{e}_j) = 0 \quad \forall j = 1, \dots, n.$$

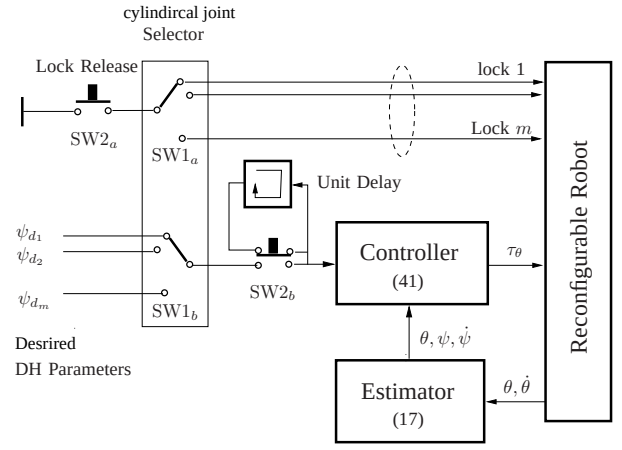


Fig. 5. Control architecture

TABLE II
PARAMETERS OF THE RECONFIGURABLE ROBOT

Link 1 & 2	Link 3
$d = 0.2$ m	-
$1 < L < 2$ m	$L = 0.7$ m
$-180 < \alpha^\dagger < 180$ deg	$\alpha^\dagger = 0$ deg
$\rho_x^\dagger = 0.5$ m	$\rho_x = 0.15$ m
$\rho_y^\dagger = \rho_z^\dagger = 0$	$\rho_y = \rho_z = 0$
$m^\dagger = 1$ kg	$m = 0.3$ kg
$I_{xx}^\dagger = 0.0025$ kgm ²	$I_{xx} = 0.00075$ kgm ²
$I_{yy}^\dagger = I_{zz}^\dagger = 0.0833$ kgm ²	$I_{yy} = I_{zz} = 0.0022$ kgm ²

$\dagger$ Twist angle

$\ddagger$ Quantities associated with each segment of cylindrical links

V. SIMULATION

In this section, we resort to a simulation to demonstrate that the proposed controller can indeed realize the configuration change of a robot as depicted in Fig. 1. The robot has three active joints and two lockable cylindrical joints, of which only one is released at a time. The kinematic and inertial parameters of the robot used in the simulation are given in Table II, and the controller gains are set to $K_p = \text{diag}([5, 0.8])$ and $K_d = \text{diag}([6, 1])$.

The configuration change takes place in two stages:

- (i) the first link is twisted -90 deg and its length is increased by 0.5 m, as shown in Figs. 1(b)–1(e), and
- (ii) the second link is twisted by 90 deg and its length is increased by 0.5 m, as shown in Figs. 1(f)–1(g).

The two above stages can be described by

$$\begin{array}{l} \text{1st Cyl. Joint} \\ \text{2nd Cyl. Joint} \end{array} \begin{bmatrix} 1.1 \text{ m} \\ 0^\circ \\ 1.1 \text{ m} \\ 90^\circ \end{bmatrix} \xrightarrow[\text{stage}]{} \begin{bmatrix} 1.6 \text{ m} \\ -90^\circ \\ 1.1 \text{ m} \\ 90^\circ \end{bmatrix} \xrightarrow[\text{stage}]{} \begin{bmatrix} 1.6 \text{ m} \\ -90^\circ \\ 1.6 \text{ m} \\ 180^\circ \end{bmatrix}.$$

Therefore, the desired values of the passive joints are set to

$$\psi_{d1} = \begin{bmatrix} 1.6 \text{ m} \\ -90^\circ \end{bmatrix} \quad \text{and} \quad \psi_{d2} = \begin{bmatrix} 1.6 \text{ m} \\ 180^\circ \end{bmatrix}.$$

In free space, when all cylindrical joints are locked, as explained in Section II, a conventional controller drives the

active joints to achieve the desired values

$$\theta_d = \begin{bmatrix} 26.32 \\ -35.76 \\ 16.60 \end{bmatrix} \text{ deg}$$

in order to grasp a point at location

$$r_0 = r(\theta_0, \psi_0) = \begin{bmatrix} 2.7 \\ 0.0 \\ 0.4 \end{bmatrix} \text{ m},$$

where function $r(\theta, \psi) \in \mathbb{R}^3$ represents the position of the robot EE, with θ_0 and ψ_0 being the initial joint angles and DH parameters. It turns out that forming a closed chain at this point allows both configuration changes to be performed consecutively without running into any singularity. Therefore, the constraint can be expressed by

$$\phi(\theta, \psi) = r(\theta, \psi) - r(\theta_0, \psi_0) = 0$$

The projection matrix required to implement the controller is calculated from the above kinematic function.

A. Implementation of the Reconfiguration Control

Figure 5 schematically illustrates the reconfiguration control as implemented on Matlab/Simulink. Since the reconfiguration maneuver of this particular robot has to take place in two stages, we need a mechanism to perform the reconfiguration sequentially. Selector switch SW1 determines which cylindrical joint undergoes reconfiguration, by selecting the corresponding lock and the desired set of DH parameters. Pressing switch SW2 releases the selected lock and, at the same time, sends the selected desired values to the controller. As shown in the figure, the dynamic estimator takes the values of the active joint angles and their rates and subsequently estimates the entire states. After convergence of the estimator, the states vector along with the desired values of a particular cylindrical joint are fed to the controller. Subsequently, the controller issues motor torques τ_θ in order to regulate the values of the cylindrical joint according to the desired values.

B. Results

The time histories of the passive joints, active joints, joint velocities, and joint torques during the configuration change are illustrated in Figs. 6 and 7. Note that during the time interval $t \in [0, 10]$ sec., the first cylindrical joint is released while the second one is locked to permit the first stage of the reconfiguration. The second stage takes place at $t \in (11, 21]$ sec., where the lock-release condition is reversed. The transition from the first stage of the reconfiguration to the second one takes place at $t \in [10, 11]$ sec. As can be seen from the time histories of the link lengths and the twist angles in Figs. 6(a) and 6(b), the controllers were able to reconfigure the robot as was intended, and the designated kinematic parameters are changed to their desired values. The robot reconfiguration procedure is illustrated in Figs. 1(b)–1(g) at $t = 0, 0.8, 1.8, 10, 10.8, 20$ seconds, respectively. Figures 1(a) and 1(h) show the manipulator before and after its reconfiguration, respectively.

It is worth mentioning that, since the constrained robot loses all of its degrees of freedom during the lock-lock phase, the controller is no longer needed and the motor torque can be set to any value. Nevertheless, to avoid the discontinuity in motor torque, it can be assigned according to a continuous trajectory, say a linear one, which connects the boundary points of the lock-lock phase.

VI. CONCLUSION

We introduced a new design of reconfigurable robots with passive joints, which allow changing the robot DH parameters. Such a robot has a simpler design compared to the conventional reconfigurable robots, which are based on modular joints and a complex docking mechanism. Moreover, because it is not required to connect or disconnect the joints and the links, the configuration change can be performed reliably and autonomously.

The kinematics and control of the robot in a kinematic-constraint loop, where configuration changes take place, were thoroughly investigated. The problem of configuration change was shown to be closely related to the regulation of a constrained mechanical system with passive joints. Using the notion of linear projection operators, we developed a Lyapunov-based controller that sends torque commands to the active joints so as to take the released passive joints to their desired values. Notably, the torque command is optimized to reduce the actuation effort. The controller requires only the models of the constraint kinematics and the links gravity, but the other elements of the dynamics model such as the inertia matrix is not required. That makes the controller easy to implement and suitable for reconfiguration control, as the variation of the dynamic properties of the robot due to reconfiguration should otherwise be taken into account.

The reconfigurability conditions, under which the passive cylindrical joints become controllable and observable, have been derived. It was proven that satisfying these conditions reduces to have (i) A_ψ full rank and (ii) $\mathcal{R}(A_\psi) \subseteq \mathcal{R}(A_\theta)$. Subsequently, we showed that for the common case of the ball-joint type constraint, the first condition is fulfilled if the fixed-point on the robot EE does not lie on the axis of the cylindrical joint. The relation between the second condition and the robot axes are discussed that can be helpful to identify the attainable and unattainable configurations. To this end, for a given robot with n active joints, we showed all possible combinations in the number of the released cylindrical joints, in relation to the type of the constraint imposed on the robot end-effector to ascertain a complete configuration change.

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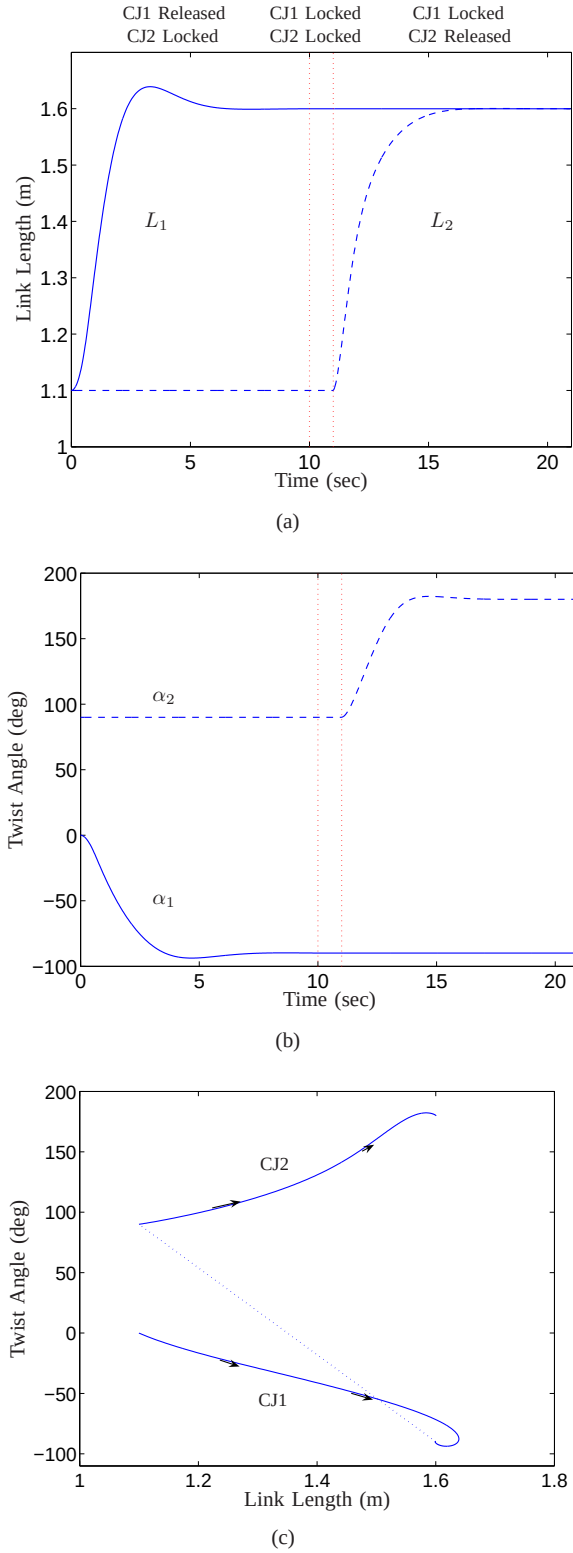


Fig. 6. Trajectories of variables associated with the passive joints during the configuration change

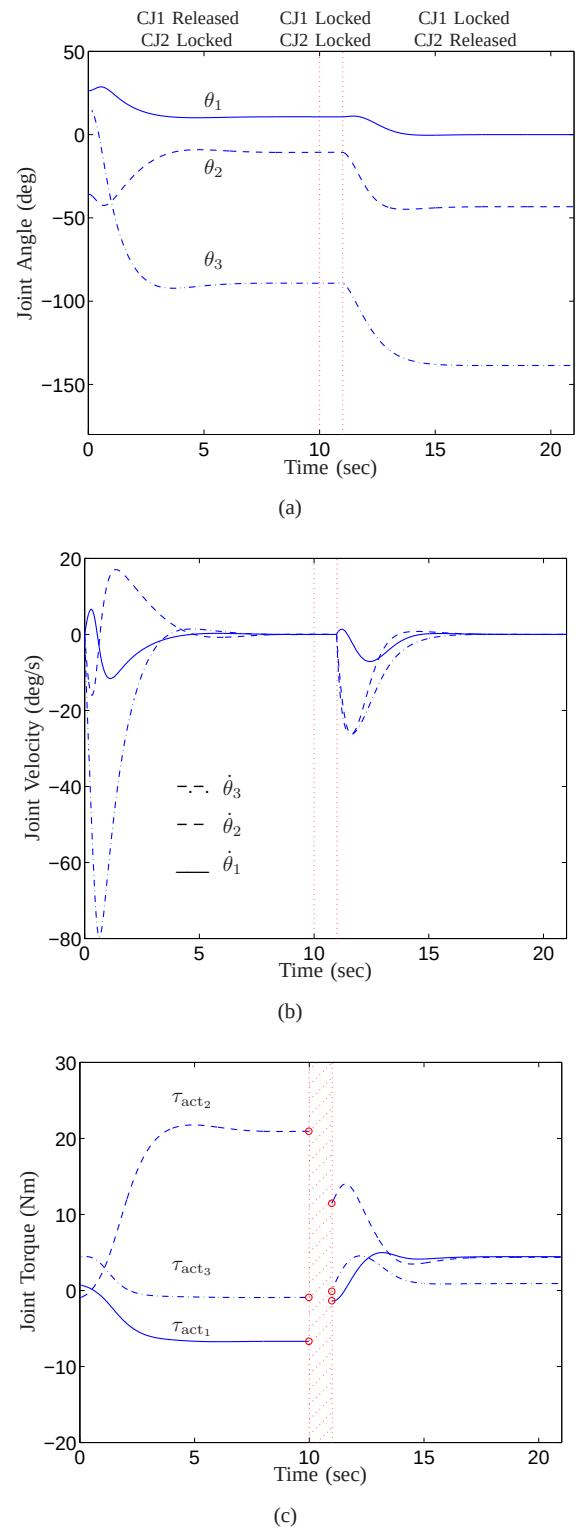


Fig. 7. Trajectories of variables associated with the active joint during the configuration change

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